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14.8 Do Two-Dimensional Distributions Differ?

We here discuss a useful generalization of the K–S test (§14.3) to *two-dimensional* distributions. This generalization is due to Fasano and Franceschini [1], a variant on an earlier idea due to Peacock [2].

In a two-dimensional distribution, each data point is characterized by an (x, y) pair of values. An example near to our hearts is that each of the 19 neutrinos that were detected from Supernova 1987A is characterized by a time t_i and by an energy E_i (see [3]). We might wish to know whether these measured pairs (t_i, E_i) , $i = 0 \dots 18$ are consistent with a theoretical model that predicts neutrino flux as a function of both time and energy — that is, a two-dimensional probability distribution in the (x, y) [here, (t, E)] plane. That would be a one-sample test. Or, given two sets of neutrino detections, from two comparable detectors, we might want to know whether they are compatible with each other, a two-sample test.

In the spirit of the tried-and-true one-dimensional K–S test, we want to range over the (x, y) -plane in search of some kind of maximum *cumulative* difference between two two-dimensional distributions. Unfortunately, cumulative probability distribution is not well-defined in more than one dimension! Peacock's insight was that a good surrogate is the *integrated probability in each of four natural quadrants* around a given point (x_i, y_i) , namely the total probabilities (or fraction of data) in $(x > x_i, y > y_i)$, $(x < x_i, y > y_i)$, $(x < x_i, y < y_i)$, $(x > x_i, y < y_i)$. The two-dimensional K–S statistic D is now taken to be the maximum difference (ranging both over data points and over quadrants) of the corresponding integrated probabilities. When comparing two data sets, the value of D may depend on which data set is ranged over. In that case, define an effective D as the average of the two values obtained. If you are confused at this point about the exact definition of D , don't fret; the accompanying computer routines amount to a precise algorithmic definition.

Figure 14.8.1 gives a feeling for what is going on. The 65 triangles and 35 squares seem to have somewhat different distributions in the plane. The dotted lines are centered on the triangle that maximizes the D statistic; the maximum occurs in the upper-left quadrant. That quadrant contains only 0.12 of all the triangles, but it contains 0.56 of all the squares. The value of D is thus 0.44. Is this statistically significant?

Even for fixed sample sizes, it is unfortunately not rigorously true that the distribution of D in the null hypothesis is independent of the shape of the two-dimensional distribution. In this respect the two-dimensional K–S test is not as natural as its one-dimensional parent. However, extensive Monte Carlo integrations have shown that the distribution of the two-dimensional D is *very nearly* identical for even quite different distributions, as long as they have the same coefficient of correlation r , defined in the usual way by equation (14.5.1). In their paper, Fasano and Franceschini tabulate Monte Carlo results for (what amounts to) the distribution of D as a function

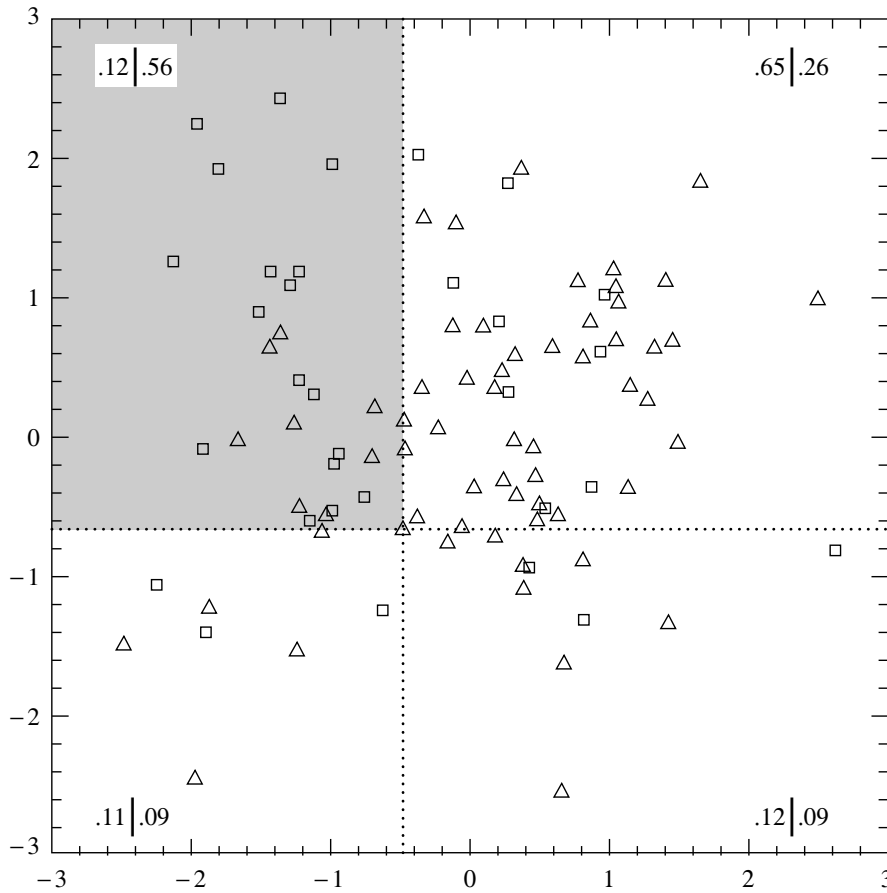


Figure 14.8.1. Two-dimensional distributions of 65 triangles and 35 squares. The two-dimensional K–S test finds that point one of whose quadrants (shown by dotted lines) maximizes the difference between fraction of triangles and fraction of squares. Then, equation (14.8.1) indicates whether the difference is statistically significant, i.e., whether the triangles and squares must have different underlying distributions.

of (of course) D , sample size N , and coefficient of correlation r . Analyzing their results, one finds that the significance levels for the two-dimensional K–S test can be summarized by the simple, though approximate, formulas

$$\text{Probability } (D > \text{observed}) = Q_{KS} \left(\frac{\sqrt{N} D}{1 + \sqrt{1 - r^2} (0.25 - 0.75/\sqrt{N})} \right) \quad (14.8.1)$$

for the one-sample case, and the same for the two-sample case, but with

$$N = \frac{N_1 N_2}{N_1 + N_2}. \quad (14.8.2)$$

The above formulas are accurate enough when $N \gtrsim 20$, and when the indicated probability (significance level) is less than (more significant than) 0.20 or so. When the indicated probability is > 0.20 , its value may not be accurate, but the

implication that the data and model (or two data sets) are not significantly different is certainly correct. Notice that in the limit of $r \rightarrow 1$ (perfect correlation), equations (14.8.1) and (14.8.2) reduce to equations (14.3.18) and (14.3.19): The two-dimensional data lie on a perfect straight line, and the two-dimensional K–S test becomes a one-dimensional K–S test.

The significance level for the data in Figure 14.8.1, by the way, is about 0.001. This establishes to a near-certainty that the triangles and squares were drawn from different distributions. (As in fact they were.)

Of course, if you do not want to rely on the Monte Carlo experiments embodied in equation (14.8.1), you can do your own: Generate a lot of synthetic data sets from your model, each one with the same number of points as the real data set. Compute D for each synthetic data set, using the accompanying computer routines (but ignoring their calculated probabilities), and count what fraction of the time these synthetic D 's exceed the D from the real data. That fraction is your significance.

One disadvantage of the two-dimensional tests, by comparison with their one-dimensional progenitors, is that the two-dimensional tests require of order N^2 operations: Two nested loops of order N take the place of an $N \log N$ sort. For desktop computers, this restricts the usefulness of the tests to N less than several thousand.

We now give computer implementations. The one-sample case is embodied in the routine `ks2d1s` (that is, two dimensions, one sample). This routine calls a straightforward utility routine `quadct` to count points in the four quadrants, and it calls a user-supplied routine `quadv1` that must be capable of returning the integrated probability of an analytic model in each of four quadrants around an arbitrary (x, y) point. A trivial sample `quadv1` is shown; realistic `quadv1`s can be quite complicated, often incorporating numerical quadratures over analytic two-dimensional distributions.

`kstests_2d.h`

```
void ks2d1s(VecDoub_I &x1, VecDoub_I &y1, void quadv1(const Doub, const Doub,
    Doub &, Doub &, Doub &, Doub &), Doub &d1, Doub &prob)
```

Two-dimensional Kolmogorov-Smirnov test of one sample against a model. Given the x and y coordinates of $n1$ data points in arrays `x1[0..n1-1]` and `y1[0..n1-1]`, and given a user-supplied function `quadv1` that exemplifies the model, this routine returns the two-dimensional K-S statistic as `d1`, and its p -value as `prob`. Small values of `prob` show that the sample is significantly different from the model. Note that the test is slightly distribution-dependent, so `prob` is only an estimate.

```
{
    Int j,n1=x1.size();
    Doub dum,dumm,fa,fb,fc,fd,ga,gb,gc,gd,r1,rr,sqen;
    KSdist ks;
    d1=0.0;
    for (j=0;j<n1;j++) {
        Loop over the data points.
        quadct(x1[j],y1[j],x1,y1,fa,fb,fc,fd);
        quadv1(x1[j],y1[j],ga,gb,gc,gd);
        if (fa > ga) fa += 1.0/n1;
        if (fb > gb) fb += 1.0/n1;
        if (fc > gc) fc += 1.0/n1;
        if (fd > gd) fd += 1.0/n1;
        d1=MAX(d1,abs(fa-ga));
        d1=MAX(d1,abs(fb-gb));
        d1=MAX(d1,abs(fc-gc));
        d1=MAX(d1,abs(fd-gd));
        For both the sample and the model, the distribution is integrated in each of four
        quadrants, and the maximum difference is saved.
    }
    pearsn(x1,y1,r1,dum,dumm);
    Get the linear correlation coefficient r1.
```

```

sqen=sqrt(Doub(n1));
rr=sqrt(1.0-r1*r1);
Estimate the probability using the K-S probability function.
prob=ks.qks(d1*sqen/(1.0+rr*(0.25-0.75/sqen)));
}

```

```

void quadct(const Doub x, const Doub y, VecDoub_I &xx, VecDoub_I &yy, Doub &fa,
Doub &fb, Doub &fc, Doub &fd)

```

kstests_2d.h

Given an origin (x,y), and an array of nn points with coordinates xx[0..nn-1] and yy[0..nn-1], count how many of them are in each quadrant around the origin, and return the normalized fractions. Quadrants are labeled alphabetically, counterclockwise from the upper right. Used by ks2d1s and ks2d2s.

```

{
    Int k,na,nb,nc,nd,nn=xx.size();
    Doub ff;
    na=nb=nc=nd=0;
    for (k=0;k<nn;k++) {
        if (yy[k] == y && xx[k] == x) continue;
        if (yy[k] > y)
            xx[k] > x ? ++na : ++nb;
        else
            xx[k] > x ? ++nd : ++nc;
    }
    ff=1.0/nn;
    fa=ff*na;
    fb=ff*nb;
    fc=ff*nc;
    fd=ff*nd;
}

```

```

void quadvl(const Doub x, const Doub y, Doub &fa, Doub &fb, Doub &fc, Doub &fd)

```

quadvl.h

This is a sample of a user-supplied routine to be used with ks2d1s. In this case, the model distribution is uniform inside the square $-1 < x < 1$, $-1 < y < 1$. In general, this routine should return, for any point (x,y), the fraction of the total distribution in each of the four quadrants around that point. The fractions, fa, fb, fc, and fd, must add up to 1. Quadrants are alphabetical, counterclockwise from the upper right.

```

{
    Doub qa,qb,qc,qd;
    qa=MIN(2.0,MAX(0.0,1.0-x));
    qb=MIN(2.0,MAX(0.0,1.0-y));
    qc=MIN(2.0,MAX(0.0,x+1.0));
    qd=MIN(2.0,MAX(0.0,y+1.0));
    fa=0.25*qa*qb;
    fb=0.25*qb*qc;
    fc=0.25*qc*qd;
    fd=0.25*qd*qa;
}

```

The routine ks2d2s is the two-sample case of the two-dimensional K-S test. It also calls quadct, pearsn, and KSdist::qks. Being a two-sample test, it does not need an analytic model.

```

void ks2d2s(VecDoub_I &x1, VecDoub_I &y1, VecDoub_I &x2, VecDoub_I &y2, Doub &d,
Doub &prob)

```

kstests_2d.h

Two-dimensional Kolmogorov-Smirnov test on two samples. Given the x and y coordinates of the first sample as n1 values in arrays x1[0..n1-1] and y1[0..n1-1], and likewise for the second sample, n2 values in arrays x2 and y2, this routine returns the two-dimensional, two-sample K-S statistic as d, and its p-value as prob. Small values of prob show that the two samples are significantly different. Note that the test is slightly distribution-dependent, so prob is only an estimate.